

Virtual Operating System for Windows to Linux Migration

Ms. N.H.Giri

Department of Computer Engineering
Universal College of Engineering & Research, Pune, India.
neelam0302@gmail.com

Mr. V.N.Nandgaonkar

Department of Computer Engineering
Universal College of Engineering and Research, Pune, India.
vikas.nandgaonkar@gmail.com

Mr. Rahul Gosavi

Department of Computer Engineering
Universal College of Engineering and Research, Pune, India.
facebook8657577469@gmail.com

Abstract— The Windows is most popular operating system in the market. But it is not more secure than Linux, it is not high reliable, it does not provides hardware flexibility and better choices, it does not reliable. But it is more user friendly operating system due to its attractive GUI. Different software programs are designed for different operating systems, and most will not work on systems that they were designed. So many of the companies are migrated from Windows to Linux. So this paper examines the survey of Windows to Linux migration with wine. Wine is re-implementation of Windows API. It also has a unique architecture. It implements as a compatibility layer over a Unix-like kernel which allows Unix-like operating system to run 32 bit application of Windows with wine and its libraries. Organizations consider migrating to Linux to reduce costs and increase and improve stability, security, uptime, and availability.

Keywords— Virtual OS, Migration, Global Terminal, Windows API, Virtual Kernel, DLL, Wine.

Introduction

As the most well-liked package, especially in the main client marketplace, Windows has a lot of benefits within the market programs. Today's business level computing resources are much more powerful that they usually have excess capability. By using the virtualization hardware and allocating components of virtualization area unit is mostly based on the important requirement of users and applications. The available computing power, storage space and network information measure (bandwidth) is used more effectively. Physical computing machine will not idle or performing below their capabilities as a result of there area unit few connected users or as a result of the hosted application happens to be less hard to please than the server can handle.

Windows to Linux Migration supply software packages to developer isolated and constrained base environments. Rather than purchasing dedicated physical hardware and software package migration is done on the present hardware. Because every migrated virtual machine is freelance and isolated from all the alternative machine. Programmers can run software package while not having care regarding touch alternative applications and external elements touching the execution of their code.

I. LITERATURE SURVEY

In 2013, 'Wang-Ya-Jun' introduces the operating Linux software on Windows platform^[1]. Because they thought running Linux application programs in Windows operating system is one of the basic content of researching the interoperability technology of Linux and Windows. And now a day both operating system started a fierce competition in desktop domain, server domain and embedded domain throughout the world. But Microsoft Windows released a cost-free software toolkit SFU-Service For Unix which makes Windows compatible with Unix system operation. In this they found that the Source Code Refactoring^{[2][4]} can be used to carry on second designing of the source codes, so that source code fit in the application programming interface supported by Windows.

In 2011, 'Maohua Lu' introduces the Bootstrapped Migration^[2] for Linux OS. In that they discover a Swap Disk-Based Migration which suspend all the processes, objects which created on the physical machine and saves the context of process and object for resume that work. And after that Migration Kernel get Restore to performing all the activity and work as virtually. Also they suggest the Memory-to-Memory Migration and Efficient Device State Migration^[2].

In 2014 Daniel Mosse introduces the Concurrent Page Migration. The promotion and demotion of pages between DRAM and PCM requires copying pages from one type of memory to the other. This procedure is termed page migration and it is controlled by the OS.

Wine is not an emulator is open source implementation of the Windows API on top of the Linux. It's approach is to re-implement the core of Windows API in Unix-like operating system. In order to allow running Windows application natively. This approach has appeal to advocates of open-source since a user does not need to purchase a copy of Windows as opposed to emulation or virtualization. Wine rests in the fact that the development process has been difficult due to lack of detailed documentation on the Windows API and the necessity of Wine to develop many complicated APIs such as Direct3D^[3] and DirectDraw on top of the core kernel. Wine executable runs as just a normal executable and the inter-process communication service is provided by the Wine server. Wine completely re-

implements all of the Core System DLLs, including Windows DLL, GDI32, User32, and Kernel32 DLL^[3].

PlayOnLinux^[7] is a piece of software that permits us to simply install and use various games and apps designed to execute with Microsoft Windows.^[14] PlayOnLinux brings a cost-free, accessible and efficient answer to this drawbacks.

- You don't need to purchase a Windows license to use PlayOnLinux.
- It is based on Wine, and so profits from all its characteristics however it keeps the user from having to contribute its quality.
- PlayOnLinux is an free open-source software.
- It uses Bash and Python Programming Structure.
- PlayOnLinux has some lack.
- Infrequently performance decrease.

II. WHY WINDOWS TO LINUX MIGRATION

Many firms have determined to migrate to UNIX based operating system for the following reasons:

A. Linux Provides Better Security and Responsibility

Security levels in Unix-based kernel generally exceeded those in Windows kernel, providing a lot of secure and manageable setting of the box with considerably more practicality in terms of security integration and repair management.^[5]

B. Linux Has Lower Total Cost of Possession (TCP)

One of the way is UNIX operating system provides a compelling money profit is that the elimination of the necessity for a Consumer Access License (CAL), which is the fee charges per consumer to access a service on a Windows server by Microsoft Windows.^[5]

C. Linux Provides Hardware Flexibility and Choice

Linux supports all the hardware platforms equipped by IBM and several different hardware vendors. Thus, enterprises are not restricted to the "Wintel" design whereas running UNIX operating system. This flexibility translates into value savings (through the employ and repurposing of hardware) and higher tractability. The flexibility of hardware is including the flexibility to run a Linux versions (from distributors like RedHat, Novell/SUSE, Debian etc.).^[5]

D. Other Reasons

Performance, availability of ERP applications, Readiness for On-Demand capability, benefits of ASCII text file.^[5]

III. PROPOSED SYSTEM

In our system, we are accepting the existing virtual kernel(Wine) for .exe file execution. So our main aim is, access

Linux and Windows operating system at the same time. And all the command of Windows platform will get executed on the Linux platform without using any software(VM Ware Workstation& Oracle Virtual Box).

There are three modules in this system:

A. Global Terminal:

In this terminal, commands of both operating system (Linux and Windows) will be executed.

B. Access the Partition of Hard Drive:

In this module, we can access the partitions of hard disk without using any virtual system(Oracle Virtual Box & VM Ware Workstation).

C. .exe Executor:

.exe Executor module will execute the .exe(32-bit) file of Windows on Linux platform with the help of Wine.

Referring to the fact that this system does not in fact emulate a x86 microprocessor architecture like other alternatives such as VirtualPC or VMWare.

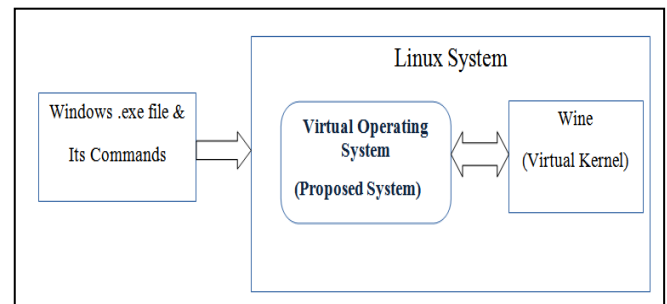


Fig. System Architecture

IV. CONCLUSION

The Linux Operating System provides some essential services for developers of system programming. This system will provide strong consistency and business oriented application program for Linux based companies. And it is more secure due to user authorization and separate shell script for every .exe file which will execute on Linux platform.

References

- [1] Wang Ya-jun: Operating Linux Software on Windows Platform, Applied Mechanics and Materials vol 394 Transtech publication(3 sept 2013)
- [2] Jui-Hao Chiang, Maohua Lu, Tzi-cker Chiueh: Bootstrapped Migration for Linux OS, ACM Computing Surveys(2011).
- [3] <https://www.winehq.org>
- [4] Yunfa Li, Wanqing Li, Congfeng Jiang: A Survey of Virtual Machine System: Current Technology and Future Trends, IEEE 3rd International Symposium on Electronic Commerce and Security.(2010)
- [5] Srirama Krishnakumar: Migration to Linux: A Roadmap for migration from windows to Linux, IBM Software Group(2005).
- [6] Balram Yadav , Vishwas Karhadkar , Yodesh Kakde : An Introduction and Study to Tiny Operating System (T.O.S.), International Journal of Advanced Research in Computer Science and Software Engineering(8 August 2013).
- [7] <https://www.playonLinux.en.uptodown.com/ubuntu>
- [8] Thomas E. Anderson, Michael D. Dahlin, Jeanna M. Neefe, David A. Patterson, Drew S. Roselli, and Randolph Y. Wang. Serverless network le systems. In Proceedings of the Fifteenth ACM Symposium on Operating Systems Principles, pages 109-126. ACM Press, December 2010.
- [9] Violeta Medina, Juan Manuel Garcia: A Survey of Migration Mechanism of Virtual Machine, ACM Computing Surveys, vol 46(Jan2014).
- [10] Jui-Hao Chiang, Maohua Lu, Tzi-cker Chiueh: Bootstrapped Migration for Linux OS, ACM Computing Surveys(2011).
- [11] Tsutomu Nomoto, Yoshihiro Oyama, Hideki Eiraku, Takahiro Shinagawa, Kazuhiko Kato: Using a Hypervisor to Migrate Runnig Operating Systems to Secure Virtual Machines, Annual IEEE 34th Computer Software and Application Conference(2010).
- [12] Franziska Roesner, Tadayoshi Kohno, Alexander Moshchuk, Bryan Parno, He-len J. Wang: User Driven Access Control: Re-thinking Permission Granting in Modern Operating System, IEEE Symposium on Security and Privacy(2012).
- [13]<https://www.digitaltrends.com/virtualization/virtualOS>
- [14] <https://www.winehq.org>
- [15] Bo Qu, Zhaozhi Wu : Design of ARM Based Embedded Operating System Micro Kernel, Proceeding on 2nd International conference on computer science and electronic Engineering (2013).